ALICE L WALTON SCHOOL OF MEDICINE

Penicillins

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Adapted from: Penicillins

Listen to this Brick

0:00 / 20:13 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Describe the mechanism of action of beta lactam antibiotics.
2. Explain the role of β-lactamase inhibitors.
3. Classify penicillins into subgroups based on their chemical structure, spectrum of activity and resistance to β-lactamases.
4. Recognize th associated with beta lactam antibiotics.

CASE CONNECTION

AL, a 60-year-old woman is brought to the hospital. She has a productive cough and is running a temperature of 103 degrees F. A left, lower lobe pneumonia was diagnosed, and she was started on oral amoxicillin. After 24 hours, her symptoms did not improve, and the cultures returned showing that *S. pneumoniae* was not sensitive to amoxicillin.

Is it possible for the *S. pneumoniae* not to be susceptible to Amoxicillin? What could be the possible reason? Could an alternative agent be used?

Consider your answer as you read, and we'll revisit AL at the end of the brick.

GO TO CONCLUSION ↓

What Are Penicillins? (LO–1)

Penicillins are broad-spectrum antibiotics, meaning they attack more than one species of bacteria. Penicillin structure consists of a fused β-lactam ring (so it's also called a β-lactam antibiotic) and a thiazolidine ring (Figure 1). An amide and a carboxylic acid group are also present. The variation is mainly limited to the side chain (R) group of the amide, but prodrugs have been developed by modifying the carboxylic acid. The chemistry is essential here since it plays a role in determining and managing hypersensitivity reactions to Penicillin. Also, in clinical practice, antibiotics are commonly referred by their chemical structures.

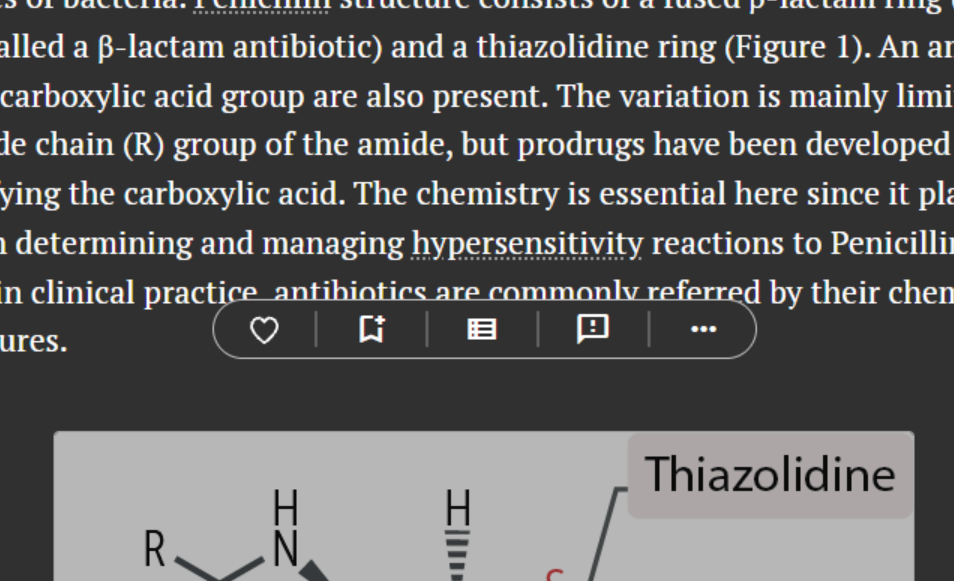


Figure 1

Credit: ©ScholarRx

What is the core chemical structure of penicillins?

Penicillin G, the natural penicillin, is broken down to make 6-aminopenicillanic acid, which is used as a precursor to make other semi-synthetic penicillins. Modifications to this structure were made to develop other penicillins. We will revisit these classes in detail later in the discussion.

What is the Mechanisms of Action of Penicillin?

The core structure of penicillins (and the other β-lactam antibiotics) is the β-lactam ring, which binds to and irreversibly (noncompetitively) inhibits the transpeptidase (or penicillin-binding protein [PBP]) enzyme, the enzyme that cross-links the linear peptidoglycan chain components of the cell wall (Figure 2). Cell wall synthesis is halted, and the bacteria die. It is important to note that both gram-positive and gram-negative organisms possess peptidoglycan in their cell walls, so imagine the broad spectrum of these β-lactam antibiotics.

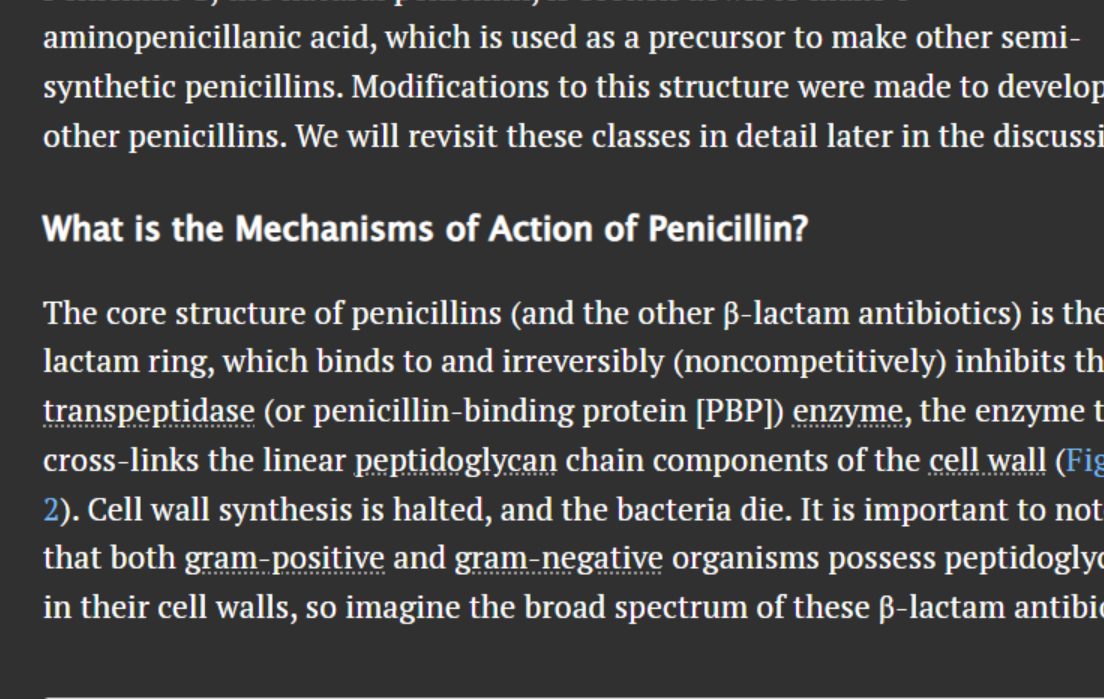


Figure 2

Now, let's turn to specific mechanisms of bactericidal action.

Cross-Link Inhibition. The β-lactam ring acts as a structural analogue for D-Ala-D-Ala, which is found in the peptidoglycan and binds to transpeptidase (or PBP), leading to blockage of peptidoglycan cross-linking in the cell wall. β-Lactam antibiotics also activate autolytic enzymes (autolysins) that cause lesions in the bacterial cell wall. Autolysins exist in the cell wall, predominantly in the gram-positive bacteria; they are normally involved with remodeling during growth and are tightly regulated. β-Lactams, in synergy with accumulated building blocks, cause activation of autolysins, which, in turn, cause rapid digestion of the cell wall. The result is osmotic death or killing of bacteria from the bactericidal effect of the antibiotic.

Efficacy on Rapid Bacterial Growth. Penicillin action is dependent on the rapid growth of bacteria in vitro and in vivo. Why? For bacteria to grow, cell walls must be built. If cell wall synthesis is inhibited (by β-lactams and glycopeptides), then bacterial growth is halted. Remember that these antibiotics are bactericidal, meaning they kill bacteria, but they can only kill the bacteria during rapid growth, which is when the bacteria are actively multiplying.

So what happens to the host cells? Penicillins do not damage host cells (e.g., of humans) because human cells neither make nor need peptidoglycan.

What Are the Pharmacokinetics of Penicillins?

Penicillins are usually excreted unchanged in the urine via glomerular filtration and tubular secretion.

Penicillins vary in their resistance to gastric acid and therefore vary in their oral bioavailability. Penicillin G is poorly absorbed orally due to breakdown by gastric acid in the stomach. Penicillin V has increased absorption due to its increased acid stability. Nafcillin has poor oral absorption. Aminopenicillins have increased stability to gastric acid.

Parenteral formulations of ampicillin, piperacillin, and ticarcillin are available for injection. They are polar compounds and are not metabolized extensively.

Which β-lactamase–resistant penicillin is mainly excreted in the bile?

Distribution, Bioava and Excretion

The plasma half-lives of most penicillins vary from 30 minutes to 1 hour. The penicillins are widely distributed in the body. Most cross the blood-brain barrier to penetrate the central nervous system but only when the meninges are inflamed. Penicillin and other organic acids are secreted rapidly from the CSF into the bloodstream by an active transport process. Probenecid competitively inhibits this transport and thus elevates the concentration of penicillin in CSF.

What Are Some Mechanisms of Bacterial Resistance? (LO#2)

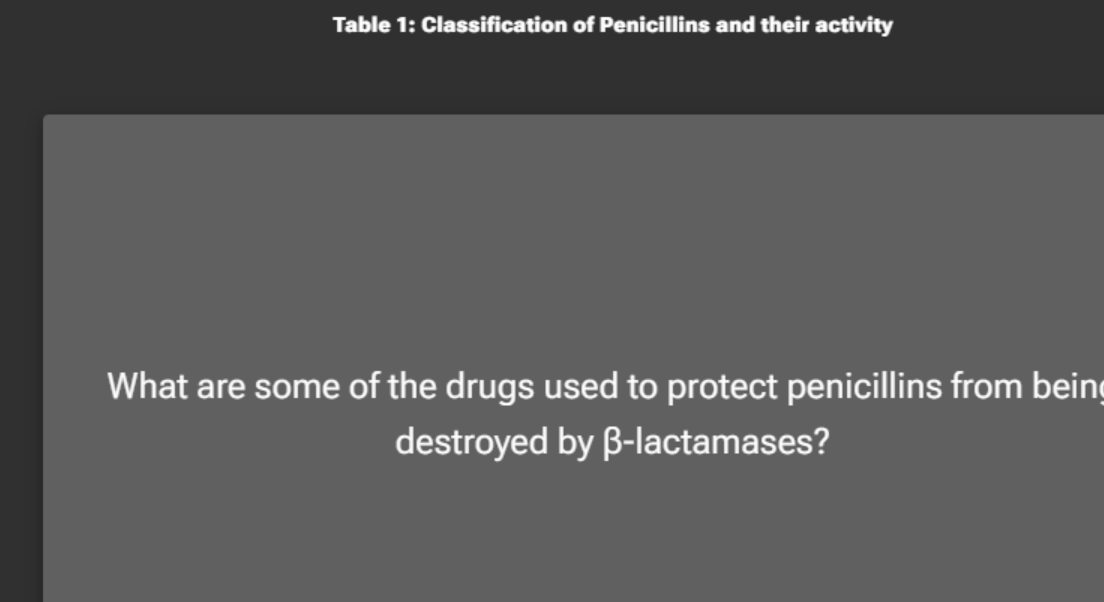
One minor mechanism of resistance is the presence of mutations in PBPs, leading to structural changes (e.g., seen in the development of resistance of *Streptococcus pneumoniae* to penicillin G). This is also responsible for methicillin resistance in staphylococci. Another minor mechanism is the change in porin structure in the outer cell wall of the membrane in some gram-negative rods (e.g., *Pseudomonas aeruginosa*), which may contribute to resistance by impeding access of penicillins to PBPs.

The major mechanism is the production of β-lactamase (penicillinase), which cleaves the β-lactam ring. Both gram-positive and gram-negative organisms produce β-lactamases. Enzymatic cleavage of the β-lactam ring results in loss of antimicrobial activity. Hence β-lactamase inhibitors were developed and co-formulated with Penicillins to "protect" the β-lactam ring. Examples of β-lactamase inhibitors include clavulanate, sulbactam, tazobactam, avibactam, and vaborbactam.

What is the main bacterial mechanism of resistance against a penicillin antibiotic?

HOW ARE PENICILLINS CLASSIFIED? (LO–3)

Penicillins are classified based on their chemical structure and spectrum of activity. The spectrum of activity, briefly described below will be discussed in detail in the organ systems for each antibiotic (Table 1).



PENICILLIN	ACTIVITY	EXAMPLES
PENICILLINS	These have the greatest activity against gram-positive organisms (gram-positive cocci, gram-positive rods). However, they have little activity against gram-negative rods, and they are susceptible to hydrolysis by β-lactamases.	Pencillin G Penicillin V
ANTISTAPHYLOCOCCAL PENICILLINS	These penicillins are resistant to staphylococcal β-lactamases. Hence, they are active against staphylococci and streptococci, but have limited spectrum of activity.	Nafcillin Oxacillin Cloxacillin
AMINOPENICILLINS	Th neg coformulated with β-lactamase inhibitors, such as clavulanate or sulbactam, which extend the activity against some β-lactamase–producing organisms.	Ampicillin Amoxicillin Ampicillin/Sulbactam Amoxicillin/Clavulanate
ANTIPSEUDOMONAL PENICILLINS	Piperacillin is a derivative of ampicillin and has a broader spectrum of activity than aminopenicillins. These drugs retain the antibacterial spectrum of penicillin and they have improved activity against gram-positive and gram-negative organisms. Coformulation of Piperacillin/Tazobactam has activity against serious Pseudomonal infections (hence the name).	Piperacillin/Tazobactam

Table 1: Classification of Penicillins and their activity

What are some of the drugs used to protect penicillins from being destroyed by β-lactamases?

What Are Some Side Effects of Penicillin Treatment? (LO–4)

Penicillin is degraded into several metabolites including benzylpenicilloyl derivatives. An immune response can be elicited by antibodies made to these metabolites, with im plicated antibody.

Hypersensitivity Reactions

A type 1 or immediate hypersensitivity reaction can occur usually within 2-20 minutes of drug administration. When IgE antibodies on mast cells make contact with the antigen (the metabolites), this interaction leads to immediate degranulation of mast cells, releasing various mediators including histamine. These mediators can lead to the classic signs and symptoms of allergy, including bronchoconstriction, urticaria (hives) in skin, hypotension, and wheezing.

Type 2 hypersensitivity is also seen with penicillins, in which IgG reacts with penicillin absorbed on red blood cells. This can lead to Coombs-positive hemolytic anemia and usually occurs in a small percentage of patients receiving increased doses of IV penicillin for a prolonged period of time (the anemia is usually reversible upon drug termination).

A type 3 hypersensitivity to penicillin can result due to circulating antibody–antigen complexes that can deposit in the skin, kidneys, and blood vessels and cause tissue damage through complement activation. A serum sickness–like syndrome (the prototype immune complex disease) can occur 1 to 3 weeks after the start of penicillin therapy. This can manifest as urticaria, diffused lymphadenopathy, and arthralgia.

Type 4 hypersensitivity reactions can also occur with penicillin exposure. This is believed to be the mechanism in which penicillin causes drug-induced tubular interstitial nephritis. It presents with fever, eosinophilia, rash, and hematuria (presence of blood in the urine).

CLINICAL CORRELATION

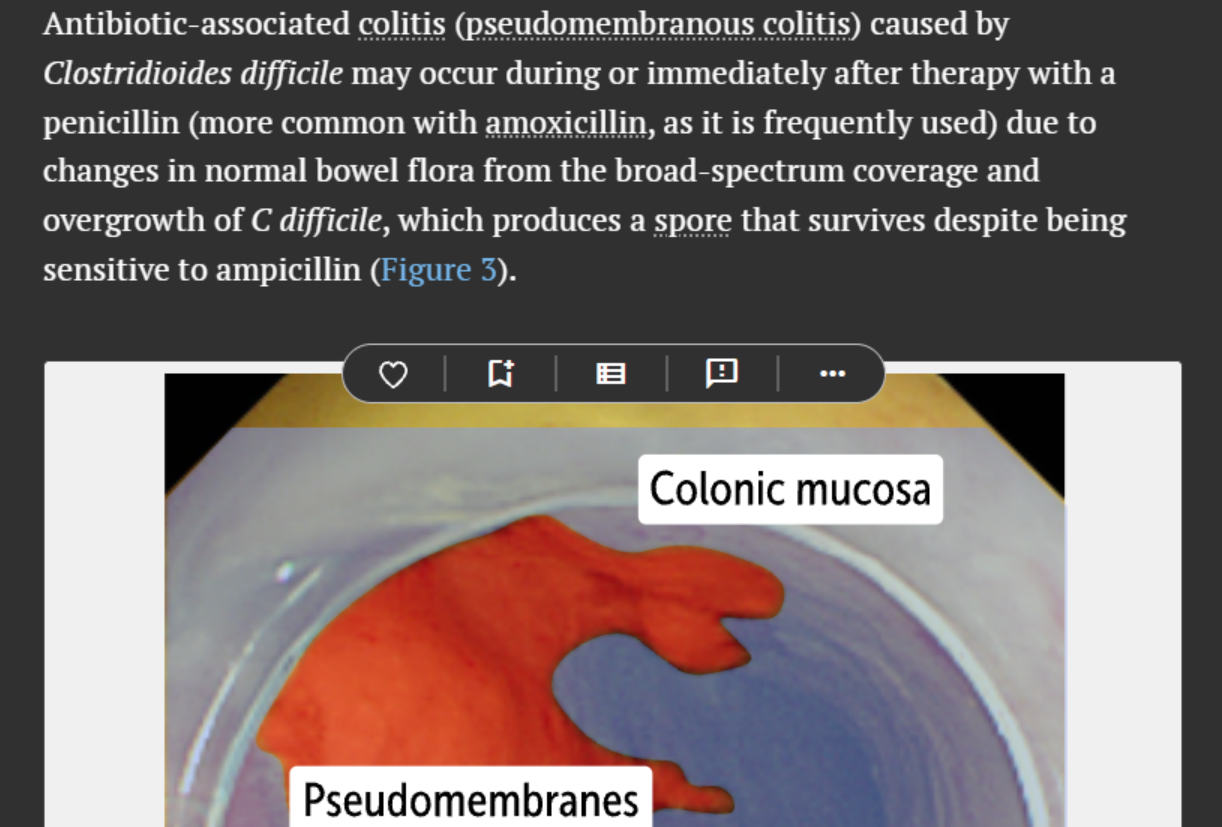
Penicillin allergy is reported by up to 10% of patients. However, more than 90% of those with a reported penicillin allergy do not exhibit IgE-mediated sensitivity when skin testing is performed. This discrepancy arises either from inappropriate labeling as allergic or from the resolution of an earlier allergy over time. Among patients with a confirmed IgE-mediated penicillin allergy, studies indicate that 97% will tolerate other β-lactams. Despite these reassuring statistics, beta-lactam drugs remain among the leading causes of drug-induced anaphylaxis.

Jarisch–Herxheimer Reaction

This occurs in patients being treated with penicillin G for a spirochete infection (usually syphilis caused by *Treponema pallidum*). It is a flu-like syndrome (fever, chills, headache) due to the release of toxins by the killed bacteria.

Gastrointestinal Effects

Nausea and diarrhea are reported with penicillins, especially ampicillin. Antibiotic-associated colitis (pseudomembranous colitis) caused by *Clostridioides difficile* may occur during or immediately after therapy with a penicillin (more common with amoxicillin, as it is frequently used) due to changes in normal bowel flora from the broad-spectrum coverage and overgrowth of *C difficile*, which produces a spore that survives despite being sensitive to ampicillin (Figure 3).



QUIZ

Figure 3

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CLINICAL CORRELATION

Pseudomembranous f inflammatory pseudomembranes in copy may reveal a red inflamed mucosa and areas of white exudate (pseudomembranes). Necrosis of the mucosal surface occurs underneath the pseudomembranes.

Skin Rash

Ampicillin and amoxicillin (which is more commonly used) have been found to cause a maculopapular skin rash consisting of small, flat, discolored spots and papules (small, raised bumps), that doesn't appear to be an allergic reaction. Patients who have infectious mononucleosis are at an increased risk of developing such a rash (Figure 4).

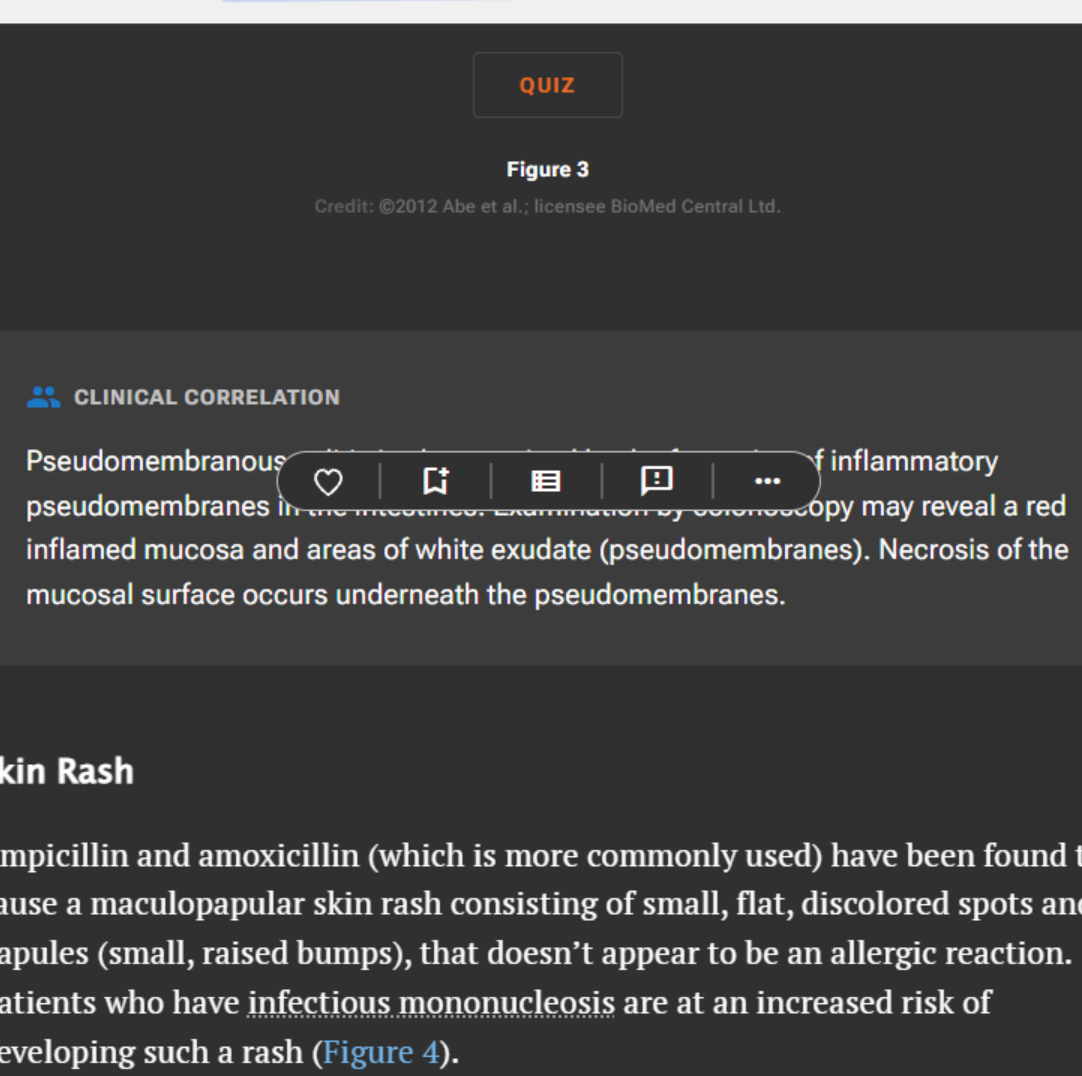
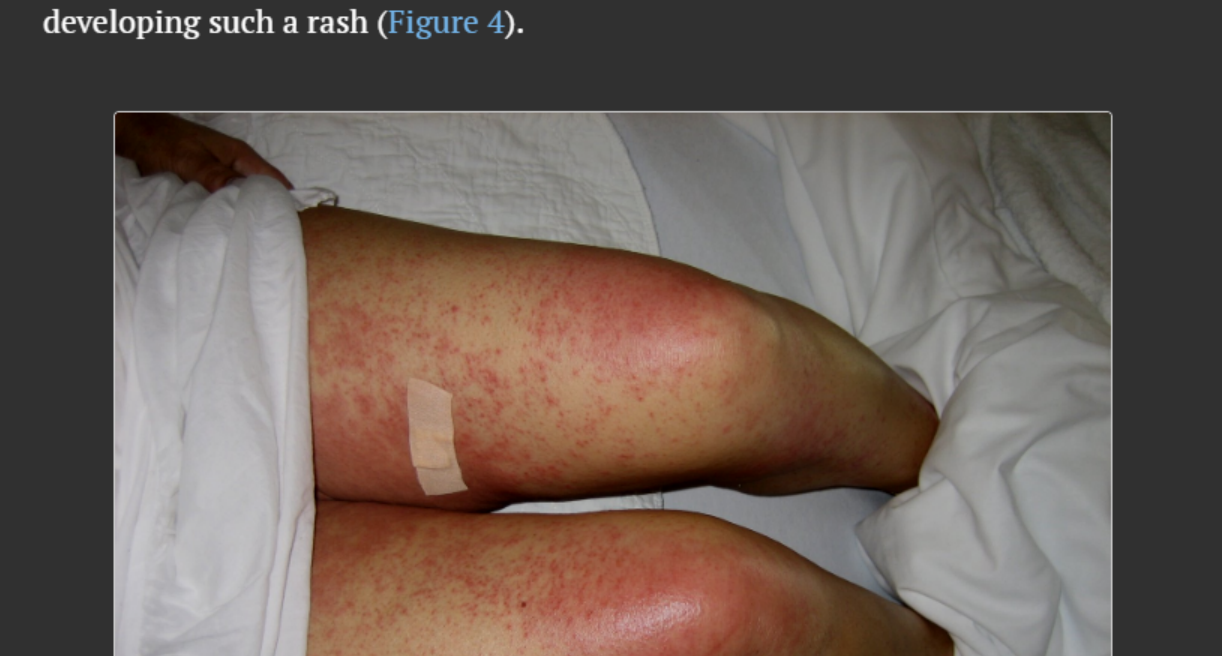


Figure 4



Adverse Drug Effect

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to AL, how do you explain the lack of susceptibility? Could we use an alternative agent?

A mechanism for bacterial resistance to Penicillin drugs is the production of β-lactamase (penicillinase) by the bacteria, which cleaves the β-lactam ring. It could be possible that the bacteria developed resistance by inactivating the aminopenicillin (am be to use products that are co-formulated with β-lactamase (penicillinase) inhibitors, such as Amoxicillin/Clavulanate or use piperacillin/tazobactam.

Summary

- Penicillins are broad-spectrum antibiotics, with a core structure consisting of a fused β-lactam ring and a thiazolidine ring.
- The β-lactam ring of penicillins binds to and irreversibly inhibits the transpeptidase enzyme, halting cell wall synthesis and leading to bacterial death.
- The primary mechanism of bacterial resistance against penicillins is the production of β-lactamase, which cleaves the β-lactam ring, rendering the antibiotic ineffective.
- Penicillins are classified based on their chemical structure and spectrum of activity, and are mainly excreted unchanged in the urine.
- β-lactamase inhibitors such as clavulanate, sulbactam, and tazobactam are co-formulated with penicillins to protect the β-lactam ring and overcome bacterial resistance.
- Common side effects of penicillin treatment include hypersensitivity reactions, gastrointestinal effects, and skin rashes.

Review Questions

1. A patient presents with a penicillin-resistant pneumococcal strain. What is the most likely mechanism for this resistance?

A. ○ Alterations in porin structure
B. ○ β-Lactamase production
C. ● Changes in the chemical structure of target penicillin-binding proteins
D. ○ Changes in the D-Ala-D-Ala building block of the peptidoglycan precursor
E. ○ Decreased intracellular accumulation of penicillin G

Hide Explanation

The correct answer is changes in the chemical structures of target penicillin-binding proteins (C). Alterations in porin structure (A) is also a mechanism of resistance but not related with pneumococcal resistance. β-Lactamase production (B) is the most common mechanism of resistance in most gram-positive and gram-negative organisms; however, *Streptococcus pneumoniae* has not been found to produce β-Lactamases. Changes in the D-Ala-D-Ala building block of the peptidoglycan precursor are the basis of the mechanism of resistance to vancomycin (D). Decreased intracellular accumulation of penicillin G (E) is also a mechanism of resistance but not related with pneumococcal resistance.

2. Which of the following is true regarding penicillin?

A. ○ Ampicillin is the drug of choice for serious *Staphylococcus aureus* infections.
B. ○ β-Lactam killing, while aminogly
C. ○ Penicillin is excreted in bile.
D. ● Penicillin is excreted unchanged in the urine.
E. ○ Penicillin undergoes enterohepatic recycling.

Hide Explanation

The correct answer is penicillin is excreted unchanged in the urine (D). Penicillins are usually excreted unchanged in the urine via glomerular filtration and tubular secretion. Ampicillin is the drug of choice for serious *Staphylococcus aureus* infections (A) is incorrect; that's nafcillin's claim to fame. β-Lactam antibiotics cause concentration-dependent killing, while aminoglycosides cause time-dependent killing (B) is incorrect, as the reverse is true. Penicillin is excreted in bile (C) is incorrect; however, nafcillin is mainly excreted in the bile. Penicillin undergoes enterohepatic recycling (E) is incorrect; it's ampicillin that undergoes enterohepatic recycling.

3. What is the primary action of β-lactam antibiotics?

A. ● Inhibition of cross-linking
B. ○ Autolysin activation
C. ○ Protein synthesis inhibition
D. ○ DNA repl
E. ○ RNA synthesis inhibition

Hide Explanation

The correct answer is inhibition of cross-linking (A). β-Lactam antibiotics inhibit the cross-linking of peptidoglycan in the bacterial cell wall by binding to transpeptidase. Autolysin activation (B) is a secondary effect that contributes to cell wall digestion. Protein synthesis inhibition (C) and DNA replication inhibition (D) are mechanisms of other antibiotic classes. RNA synthesis inhibition (E) is also not a primary action of β-lactam antibiotics.

4. Why do penicillins not damage human cells?

A. ● No peptidoglycan in human cells
B. ○ Human cells grow slowly
C. ○ Human cells lack cell walls
D. ○ Human cells have different enzymes
E. ○ Human cells are immune

Hide Explanation

The correct answer is uman cells do not have peptidoglycan, which is the target of penicillins. 'Human cells grow slowly' (B) is incorrect because the growth rate of human cells is not relevant to penicillin's mechanism. 'Human cells lack cell walls' (C) is incorrect because the absence of cell walls is not the specific reason; it's the absence of peptidoglycan. 'Human cells have different enzymes' (D) is incorrect because the specific target of penicillin is peptidoglycan, not enzymes. 'Human cells are immune' (E) is incorrect because immunity is not the reason; it's the lack of peptidoglycan.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe the mechanism of action of beta lactam antibiotics.~~
02 ☑ ~~Explain the role of β-lactamase inhibitors.~~
03 ☑ ~~Classify penicillins into subgroups based on their chemical structure, spectrum of activity and resistance to β-lactamases.~~
04 ☑ ~~Recognize the most common adverse drug reactions associated with be~~

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Objective Confidence: 4/4